

# XC6206 Series



*Low ESR Cap. Compatible Positive Voltage Regulators*

- ◆ CMOS Low Power Consumption
- ◆ Dropout Voltage : 160mV @ 100mA  
: 400mV @ 200mA
- ◆ Output Current : More Than 250mA (5.0V type)
- ◆ Highly Accurate :  $\pm 2\%$
- ◆ Output Voltage Range : 1.2V ~ 5.0V
- ◆ Low ESR Capacitor Compatible

## APPLICATIONS

- Battery powered equipment
- Reference voltage sources
- Cameras, video cameras
- Portable AV systems
- Mobile phones
- Portable games

## GENERAL DESCRIPTION

The XC6206 series are highly precise, low power consumption, high voltage, positive voltage regulators manufactured using CMOS and laser trimming technologies. The series provides large currents with a significantly small dropout voltage.

The XC6206 consists of a current limiter circuit, a driver transistor, a precision reference voltage and an error correction circuit.

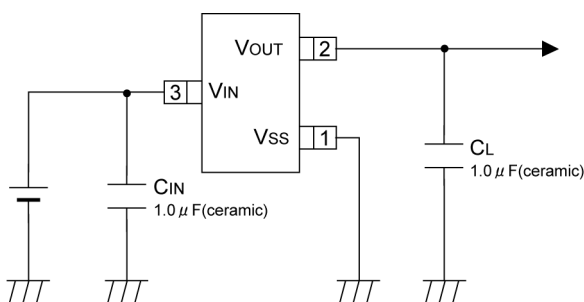
The series is compatible with low ESR ceramic capacitors. The current limiter's foldback circuit also operates as a short protect for the output current limiter and the output pin. Output voltage can be set internally by laser trimming technologies. It is selectable in 100mV increments within a range of 1.2V to 5.0V.

SOT-23, SOT-89, TO-92 and USP-6B packages are available.

## FEATURES

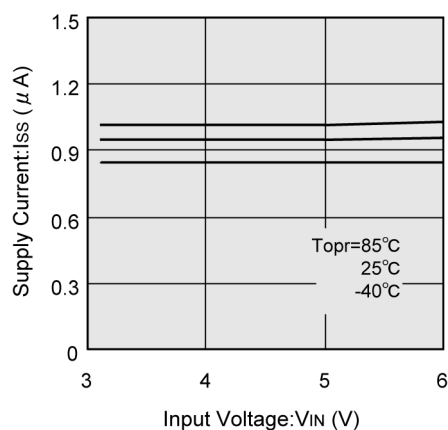
|                               |                                                                                     |
|-------------------------------|-------------------------------------------------------------------------------------|
| Maximum Output Current        | : 250mA (5.0V type)                                                                 |
| Dropout Voltage               | : 160mV @ $I_{OUT}=100mA$<br>(5.0V type)                                            |
| Maximum Operating Voltage     | : 6.0V                                                                              |
| Output Voltage Range          | : 1.2V ~ 5.0V (100mV steps)                                                         |
| Highly Accurate               | : $\pm 2\%$<br>( $\pm 30mV @ V_{OUT} < 1.5V$ )<br>( $\pm 1\% @ V_{OUT} \geq 2.0V$ ) |
| Low Power Consumption         | : $1.0\mu A$ (TYP.)                                                                 |
| Operational Temperature Range | : $-40^{\circ}C \sim 85^{\circ}C$                                                   |
| Ultra Small Package           | : SOT-23, SOT-89, TO-92<br>USP-6B                                                   |
| Low ESR Capacitor             | : Ceramic capacitor compatible                                                      |

## TYPICAL APPLICATION CIRCUIT

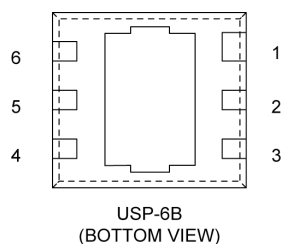
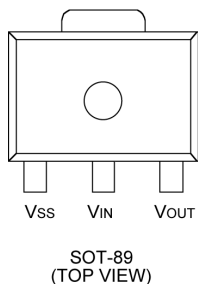
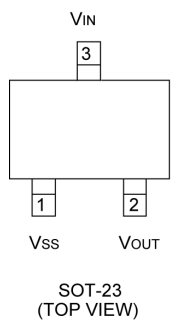


## TYPICAL PERFORMANCE CHARACTERISTICS

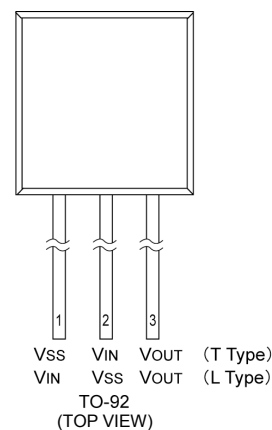
XC6206P302



## PIN CONFIGURATION



\*The dissipation pad for the USP-6B package should be solder-plated in recommended mount pattern and metal masking so as to enhance mounting strength and heat release. If the pad needs to be connected to other pins, it should be connected to the VIN pin.



## PIN ASSIGNMENT

| PIN NUMBER |                  |         |           | PIN NAME | FUNCTIONS     |
|------------|------------------|---------|-----------|----------|---------------|
| SOT-23     | SOT-89/TO-92 (T) | USP-6B  | TO-92 (L) |          |               |
| 1          | 1                | 2       | 2         | VSS      | Ground        |
| 3          | 2                | 4       | 1         | VIN      | Power Input   |
| 2          | 3                | 6       | 3         | VOUT     | Output        |
| -          | -                | 1, 3, 5 | -         | NC       | No Connection |

## PRODUCT CLASSIFICATION

### Ordering Information

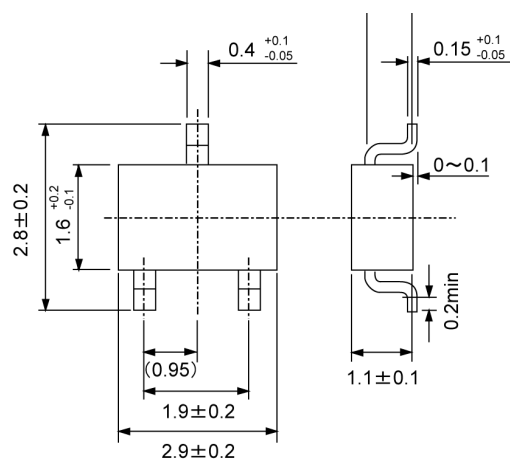
XC6206P ①②③④⑤

| DESIGNATOR | DESCRIPTION        | SYMBOL | DESCRIPTION                                                                        |
|------------|--------------------|--------|------------------------------------------------------------------------------------|
| ①②         | Output Voltage     | 12~50  | : e.g. VOUT: 3.0V→①= 3, ② = 0                                                      |
| ③          | Accuracy           | 2      | : Within $\pm 2\%$ (within $\pm 30\text{mV}$ when $V_{\text{OUT}} < 1.5\text{V}$ ) |
|            |                    | 1 *    | : Within $\pm 1\%$                                                                 |
| ④          | Package            | M      | : SOT-23                                                                           |
|            |                    | P      | : SOT-89                                                                           |
|            |                    | D      | : USP-6B                                                                           |
|            |                    | T      | : TO-92 (Standard)                                                                 |
|            |                    | L      | : TO-92 (Custom pin configuration)                                                 |
| ⑤          | Device Orientation | R      | : Embossed tape, standard feed                                                     |
|            |                    | L      | : Embossed tape, reverse feed                                                      |
|            |                    | H      | : Page type (TO-92)                                                                |
|            |                    | B      | : Bag (TO-92)                                                                      |

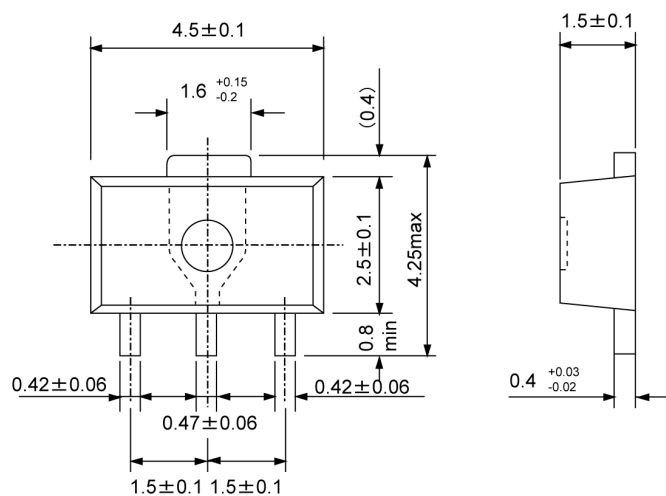
\*  $\pm 1\%$  accuracy can be set at  $V_{\text{OUT}}(T) \geq 2.0\text{V}$ .

## PACKAGING INFORMATION

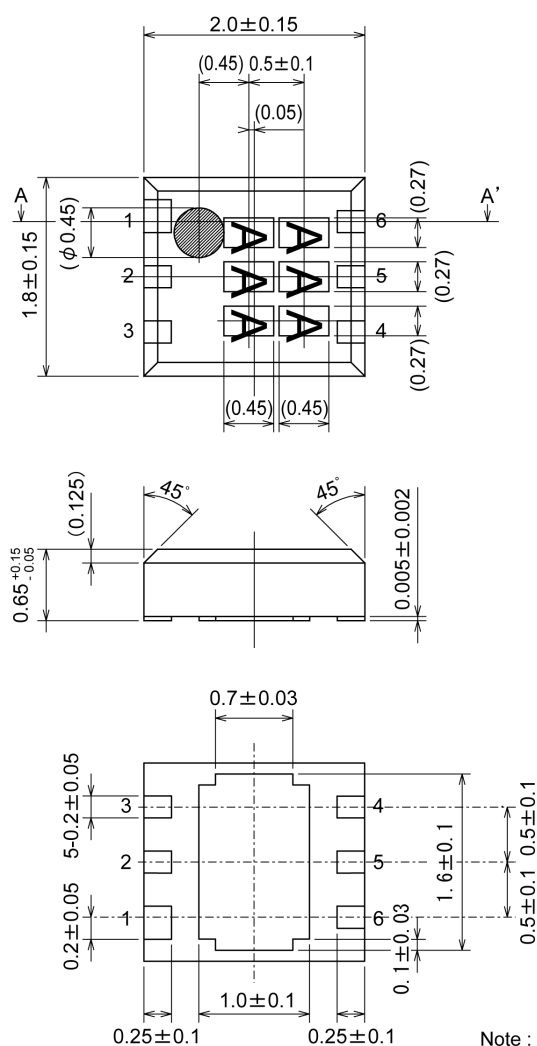
### ●SOT-23



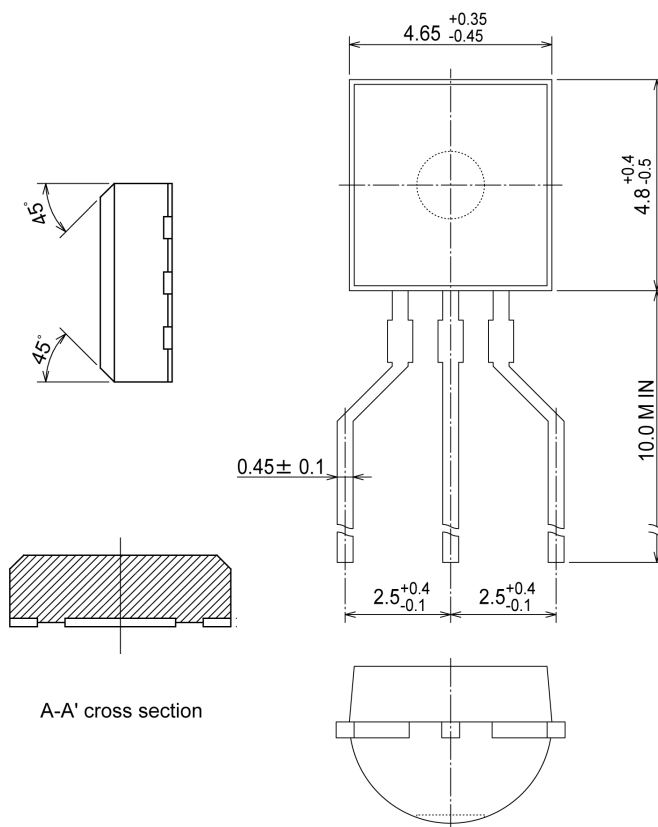
### ●SOT-89



### ●USP-6B



### ●TO-92

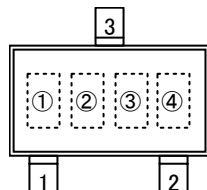


A-A' cross section

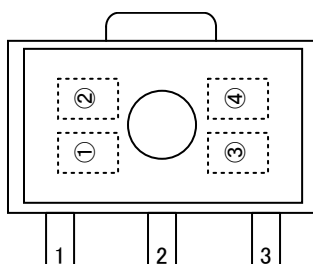
Note : Pin 1 is larger than the other pins.

## MARKING RULE

### ● SOT-23 & SOT-89



SOT-23  
(TOP VIEW)



SOT-89  
(TOP VIEW)

#### ① Represents product series

| MARK | PRODUCT SERIES |
|------|----------------|
| 6    | XC6206Pxxxxx   |

#### ② Represents three pins regulator

| MARK                   |                        | PRODUCT SERIES |
|------------------------|------------------------|----------------|
| VOLTAGE:<br>0.1 ~ 3.0V | VOLTAGE:<br>3.1 ~ 6.0V |                |
| 5                      | 6                      | XC6206Pxxxxx   |

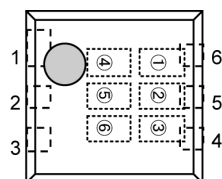
#### ③ Represents output voltage

| MARK | OUTPUT VOLTAGE (V) |     |   | MARK | OUTPUT VOLTAGE (V) |     |   |
|------|--------------------|-----|---|------|--------------------|-----|---|
| 0    | -                  | 3.1 | - | F    | 1.6                | 4.6 | - |
| 1    | -                  | 3.2 | - | H    | 1.7                | 4.7 | - |
| 2    | -                  | 3.3 | - | K    | 1.8                | 4.8 | - |
| 3    | -                  | 3.4 | - | L    | 1.9                | 4.9 | - |
| 4    | -                  | 3.5 | - | M    | 2.0                | 5.0 | - |
| 5    | -                  | 3.6 | - | N    | 2.1                | -   | - |
| 6    | -                  | 3.7 | - | P    | 2.2                | -   | - |
| 7    | -                  | 3.8 | - | R    | 2.3                | -   | - |
| 8    | -                  | 3.9 | - | S    | 2.4                | -   | - |
| 9    | -                  | 4.0 | - | T    | 2.5                | -   | - |
| A    | -                  | 4.1 | - | U    | 2.6                | -   | - |
| B    | 1.2                | 4.2 | - | V    | 2.7                | -   | - |
| C    | 1.3                | 4.3 | - | X    | 2.8                | -   | - |
| D    | 1.4                | 4.4 | - | Y    | 2.9                | -   | - |
| E    | 1.5                | 4.5 | - | Z    | 3.0                | -   | - |

#### ④ Represents production lot number

0 to 9, A to Z, reversed character of 0 to 9 and A to Z repeated.  
(G, I, J, O, Q, W excepted)

### ● USP-6B



USP-6B  
(TOP VIEW)

#### ①② Represents product series

| MARK |   | PRODUCT SERIES |
|------|---|----------------|
| ①    | ② |                |
| 0    | 6 | XC6206PxxxDx   |

#### ③ Represents three pins regulator

| MARK | TYPE                 | PRODUCT SERIES |
|------|----------------------|----------------|
| P    | Three pins regulator | XC6206PxxxDx   |

#### ④⑤ Represents output voltage

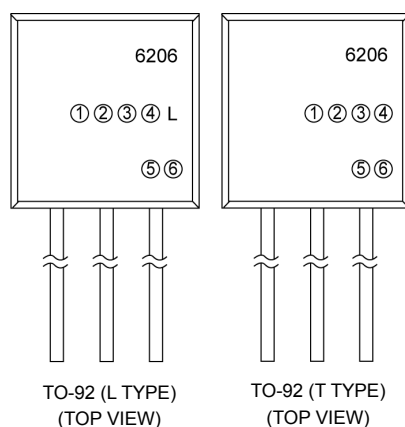
| MARK |   | OUTPUT VOLTAGE (V) | PRODUCT SERIES |
|------|---|--------------------|----------------|
| ①    | ② |                    |                |
| 3    | 3 | 3.3                | XC6206P33xDx   |
| 5    | 0 | 5.0                | XC6206P50xDx   |

#### ⑥ Represents production lot number

0 to 9, A to Z reversed (G, I, J, O, Q, W excepted)  
\* No character inversion used.

## MARKING RULE (Continued)

### TO-92



① Represents type of regulator

| MARK | PRODUCT NAME |
|------|--------------|
| P    | XC6206Pxxxxx |

②③ Represents output voltage

| MARK |   | VOLTAGE (V) | PRODUCT NAME |
|------|---|-------------|--------------|
| ②    | ③ |             |              |
| 3    | 3 | 3.3         | XC6206P33xxx |
| 5    | 0 | 3.3         | XC6206P50xxx |

④ Represents output voltage accuracy

| MARK | OUTPUT VOLTAGE ACCURACY | PRODUCT NAME |
|------|-------------------------|--------------|
| 1    | within $\pm 1\%$        | XC6206Pxx1xx |
| 2    | within $\pm 2\%$        | XC6206Pxx2xx |

⑤ Represents least significant digit of production year

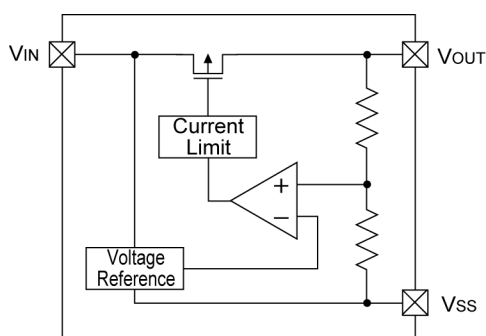
| MARK | PRODUCTION YEAR |
|------|-----------------|
| 3    | 2003            |
| 4    | 2004            |

⑥ Represents production lot number

0 to 9, A to Z repeated (G, I, J, O, Q, W excepted)

Note: No character inversion used.

## BLOCK DIAGRAM



## ■ABSOLUTE MAXIMUM RATINGS

Ta=25°C

| PARAMETER                   |        | SYMBOL           | RATINGS                                       | UNITS |
|-----------------------------|--------|------------------|-----------------------------------------------|-------|
| Input Voltage               |        | V <sub>IN</sub>  | 7.0                                           | V     |
| Output Current              |        | I <sub>OUT</sub> | 500 *                                         | mA    |
| Output Voltage              |        | V <sub>OUT</sub> | V <sub>SS</sub> - 0.3 ~ V <sub>IN</sub> + 0.3 | V     |
| Power Dissipation           | SOT-23 | P <sub>d</sub>   | 250                                           | mW    |
|                             | SOT-89 |                  | 500                                           |       |
|                             | USP-6B |                  | 100                                           |       |
|                             | TO-92  |                  | 300                                           |       |
| Operating Temperature Range |        | T <sub>opr</sub> | - 40 ~ + 85                                   | °C    |
| Storage Temperature Range   |        | T <sub>stg</sub> | - 55 ~ + 125                                  | °C    |

\* I<sub>OUT</sub>=P<sub>d</sub> / (V<sub>IN</sub>-V<sub>OUT</sub>)

## ■ELECTRICAL CHARACTERISTICS

●XC6206 series

Ta=25 °C

| PARAMETER                                  | SYMBOL                                                          | CONDITIONS                                                                                                                                                    | MIN.      | TYP.                       | MAX.      | UNITS  | CIRCUIT |
|--------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------|-----------|--------|---------|
| Output Voltage (*7)                        | V <sub>OUT(E)</sub>                                             | I <sub>OUT</sub> =30mA                                                                                                                                        | x<br>0.98 | V <sub>OUT(T)</sub><br>E-1 | x<br>1.02 | V      | ①       |
| Maximum Output Current                     | I <sub>OUTMAX</sub>                                             |                                                                                                                                                               | -         | -                          | E-2       | mA     | ①       |
| Load Regulation                            | ΔV <sub>OUT</sub>                                               | V <sub>OUT(T)</sub> >1.8V, 1mA≤I <sub>OUT</sub> ≤100mA<br>V <sub>OUT(T)</sub> ≤1.8V, 1mA≤I <sub>OUT</sub> ≤50mA                                               | -         | -                          | E-3       | mV     | ①       |
| Dropout Voltage                            | V <sub>dif1</sub>                                               | I <sub>OUT</sub> =30mA                                                                                                                                        | -         | E-4                        |           | mV     | ①       |
|                                            | V <sub>dif2</sub>                                               | V <sub>OUT(T)</sub> >1.8V: I <sub>OUT</sub> =100mA<br>V <sub>OUT(T)</sub> ≤1.8V: I <sub>OUT</sub> =60mA                                                       | -         | E-5                        |           | mV     |         |
| Supply Current                             | I <sub>DD</sub>                                                 | V <sub>CE</sub> =V <sub>IN</sub>                                                                                                                              | -         | 1.0                        | 3.0       | μA     | ②       |
| Line Regulation                            | $\frac{\Delta V_{OUT}}{\Delta V_{IN} \cdot V_{OUT}}$            | V <sub>OUT(T)</sub> <4.5V: V <sub>OUT(T)</sub> +1.0V≤V <sub>IN</sub> ≤6.0V<br>V <sub>OUT(T)</sub> ≥4.5V: 5.5V≤V <sub>IN</sub> ≤6.0V<br>I <sub>OUT</sub> =30mA | -         | 0.05                       | 0.25      | %/V    | ①       |
| Input Voltage                              | V <sub>IN</sub>                                                 |                                                                                                                                                               | 1.8       | -                          | 6.0       | V      | -       |
| Output Voltage Temperature Characteristics | $\frac{\Delta V_{OUT}}{\Delta T_{opr}} \cdot \frac{1}{V_{OUT}}$ | I <sub>OUT</sub> =30mA<br>-40 °C≤T <sub>opr</sub> ≤85 °C                                                                                                      | -         | ±100                       | -         | ppm/°C | ①       |
| Short Circuit Current                      | I <sub>short</sub>                                              | V <sub>IN</sub> =V <sub>OUT</sub> +1.5V, V <sub>OUT</sub> =V <sub>SS</sub>                                                                                    | -         | E-6                        | -         | mA     | ①       |

NOTE:

- \* 1 : V<sub>OUT(T)</sub> = Specified output voltage
- \* 2 : V<sub>OUT(E)</sub> = Effective output voltage (I.e. The output voltage when "V<sub>OUT(T)</sub>+1.0V" is provided at the V<sub>IN</sub> pin while maintaining a certain I<sub>OUT</sub> value.)
- \* 3 : V<sub>dif</sub> = {V<sub>IN</sub> 1<sup>(\*)</sup> + V<sub>OUT</sub> 1<sup>(\*)</sup>}
- \* 4 : V<sub>OUT1</sub> = A voltage equal to 98% of the output voltage whenever an amply stabilized I<sub>OUT</sub> {V<sub>OUT(T)</sub> + 1.0V} is input.
- \* 5 : V<sub>IN</sub> 1 = The input voltage when V<sub>OUT1</sub> appears as input voltage is gradually decreased.
- \* 6 : Unless otherwise stated, V<sub>IN</sub> = V<sub>OUT(T)</sub> + 1.0V
- \* 7 : When V<sub>OUT(T)</sub>≥1.5V, accuracy is ±2%.  
When V<sub>OUT(T)</sub><1.5V, accuracy is MIN.:V<sub>OUT(T)</sub>-30mV / MAX.:V<sub>OUT(T)</sub>+30mV  
+1% accuracy (MIN.: V<sub>OUT(T)</sub> x 0.99 / MAX.:V<sub>OUT(T)</sub> x 1.01) is set at V<sub>OUT(T)</sub>≥2.0V

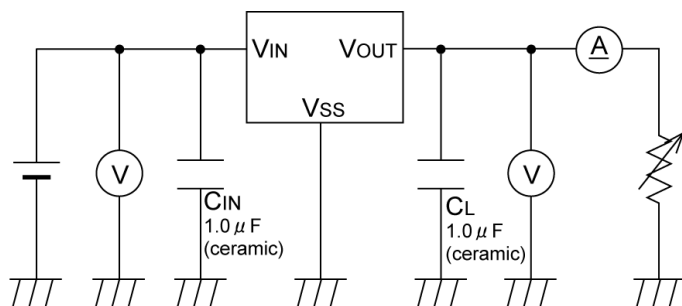
## ■ ELECTRICAL CHARACTERISTICS (Continued)

●Electrical Characteristics Chart

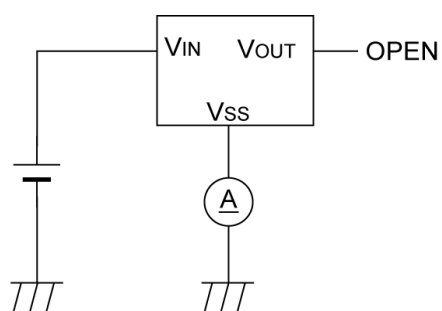
|                                     | E-1            |       |                |       | E-2                       | E-3                | E-4                  |      | E-5                  |      | E-6              |     |    |  |  |  |  |
|-------------------------------------|----------------|-------|----------------|-------|---------------------------|--------------------|----------------------|------|----------------------|------|------------------|-----|----|--|--|--|--|
| PARAMETER<br><br>SETTING<br>VOLTAGE | OUTPUT VOLTAGE |       |                |       | MAX.<br>OUTPUT<br>CURRENT | LOAD<br>REGULATION | DROPOUT<br>VOLTAGE 1 |      | DROPOUT<br>VOLTAGE 2 |      | SHORT<br>CURRENT |     |    |  |  |  |  |
|                                     | 2%<br>ACCURACY |       | 1%<br>ACCURACY |       |                           |                    | Vdif1                |      | Vdif2                |      |                  |     |    |  |  |  |  |
| VOUT(T)                             | VOUT(E) (V)    |       | VOUT(E) (V)    |       | IOUTMAX<br>(mA)           | ΔVOUT              | Vdif1                |      | Vdif2                |      | Ishort           |     |    |  |  |  |  |
|                                     | MIN.           | MAX.  | MIN.           | MAX.  | MIN.                      | MAX.               | TYP.                 | MAX. | TYP.                 | MAX. | TYP.             |     |    |  |  |  |  |
| 1.2                                 | 1.170          | 1.230 |                |       | 60                        | 40                 | 460                  | 760  | 700                  | 960  | 180              |     |    |  |  |  |  |
| 1.3                                 | 1.270          | 1.330 |                |       |                           |                    | 400                  | 650  |                      |      |                  |     |    |  |  |  |  |
| 1.4                                 | 1.370          | 1.430 |                |       |                           |                    | 350                  | 590  | 580                  | 860  |                  |     |    |  |  |  |  |
| 1.5                                 | 1.470          | 1.530 |                |       |                           | 45                 | 300                  | 510  |                      | 450  | 810              |     |    |  |  |  |  |
| 1.6                                 | 1.568          | 1.632 |                |       | 250                       |                    | 450                  | 155  |                      |      |                  |     |    |  |  |  |  |
| 1.7                                 | 1.666          | 1.734 |                |       | 200                       |                    | 410                  |      |                      |      |                  |     |    |  |  |  |  |
| 1.8                                 | 1.764          | 1.836 |                |       | 80                        |                    | 150                  | 390  |                      | 780  | 130              |     |    |  |  |  |  |
| 1.9                                 | 1.862          | 1.938 |                |       |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 2.0                                 | 1.960          | 2.040 | 1.980          | 2.020 | 120                       | 50                 | 100                  | 370  | 350                  | 710  |                  | 100 |    |  |  |  |  |
| 2.1                                 | 2.058          | 2.042 | 2.079          | 2.121 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 2.2                                 | 2.156          | 2.244 | 2.178          | 2.222 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 2.3                                 | 2.254          | 2.346 | 2.277          | 2.323 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 2.4                                 | 2.352          | 2.448 | 2.376          | 2.424 | 150                       | 55                 | 75                   | 350  | 250                  | 680  |                  |     |    |  |  |  |  |
| 2.5                                 | 2.450          | 2.550 | 2.475          | 2.525 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 2.6                                 | 2.548          | 2.652 | 2.574          | 2.626 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 2.7                                 | 2.646          | 2.754 | 2.673          | 2.727 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 2.8                                 | 2.744          | 2.856 | 2.772          | 2.828 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 2.9                                 | 2.842          | 2.958 | 2.871          | 2.929 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 3.0                                 | 2.940          | 3.060 | 2.970          | 3.030 | 200                       | 60                 | 60                   | 320  | 200                  | 630  |                  |     |    |  |  |  |  |
| 3.1                                 | 3.038          | 3.162 | 3.069          | 3.131 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 3.2                                 | 3.136          | 3.264 | 3.168          | 3.232 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 3.3                                 | 3.234          | 3.366 | 3.267          | 3.333 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 3.4                                 | 3.332          | 3.468 | 3.366          | 3.434 |                           | 65                 |                      |      |                      |      |                  |     |    |  |  |  |  |
| 3.5                                 | 3.430          | 3.570 | 3.465          | 3.535 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 3.6                                 | 3.528          | 3.672 | 3.564          | 3.636 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 3.7                                 | 3.626          | 3.774 | 3.663          | 3.737 |                           | 250                |                      |      |                      |      |                  |     | 70 |  |  |  |  |
| 3.8                                 | 3.724          | 3.876 | 3.762          | 3.838 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 3.9                                 | 3.822          | 3.978 | 3.861          | 3.939 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 4.0                                 | 3.920          | 4.080 | 3.960          | 4.040 | 75                        |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 4.1                                 | 4.018          | 4.182 | 4.059          | 4.141 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 4.2                                 | 4.116          | 4.284 | 4.158          | 4.242 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 4.3                                 | 4.214          | 4.386 | 4.257          | 4.343 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 4.4                                 | 4.312          | 4.488 | 4.356          | 4.444 | 80                        |                    | 50                   | 290  | 175                  | 600  |                  |     |    |  |  |  |  |
| 4.5                                 | 4.410          | 4.590 | 4.455          | 4.545 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 4.6                                 | 4.508          | 4.692 | 4.554          | 4.646 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 4.7                                 | 4.606          | 4.794 | 4.653          | 4.747 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 4.8                                 | 4.704          | 4.896 | 4.752          | 4.848 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 4.9                                 | 4.802          | 4.998 | 4.851          | 4.949 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |
| 5.0                                 | 4.900          | 5.100 | 4.950          | 5.050 |                           |                    |                      |      |                      |      |                  |     |    |  |  |  |  |

## ■ TEST CIRCUITS

Circuit ①



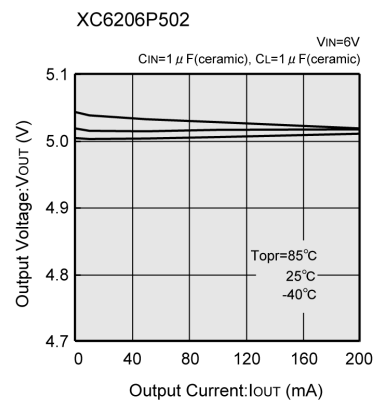
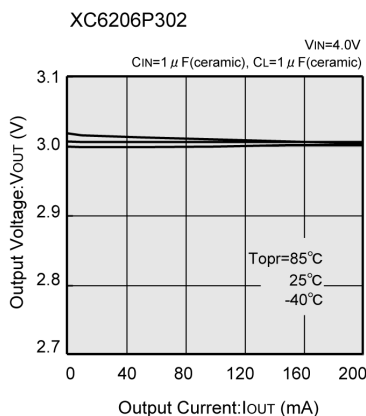
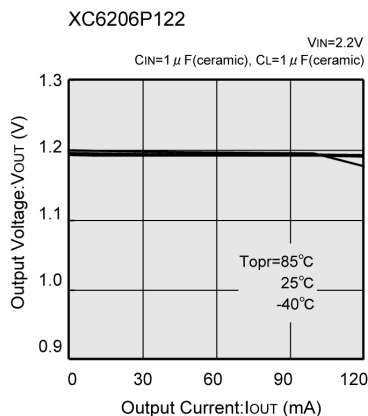
Circuit ②



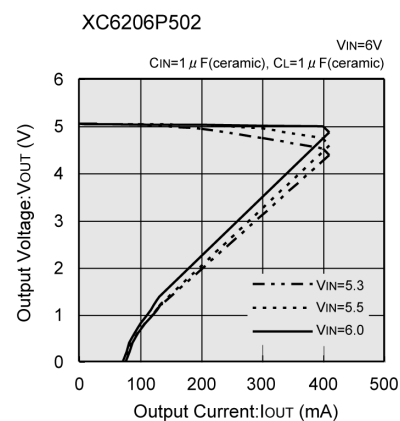
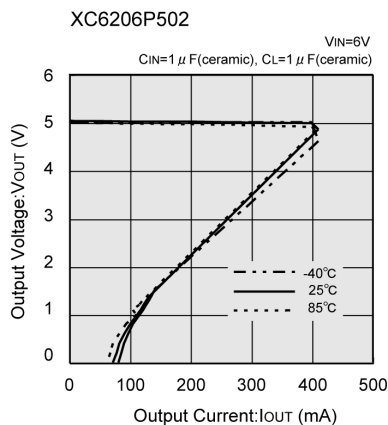
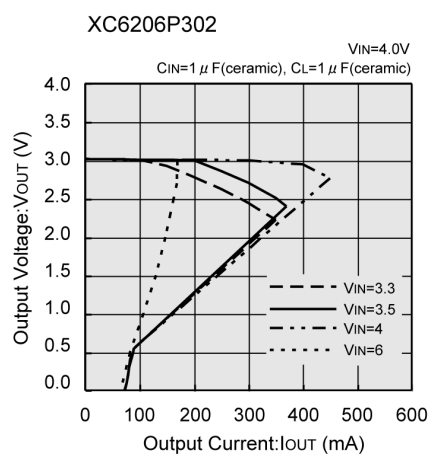
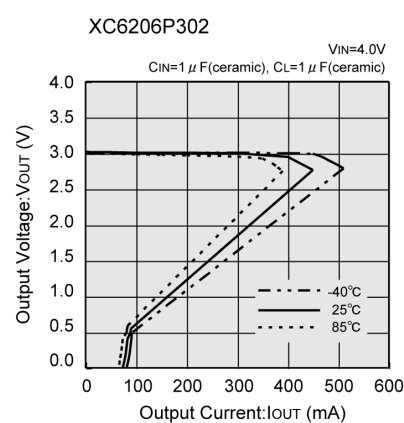
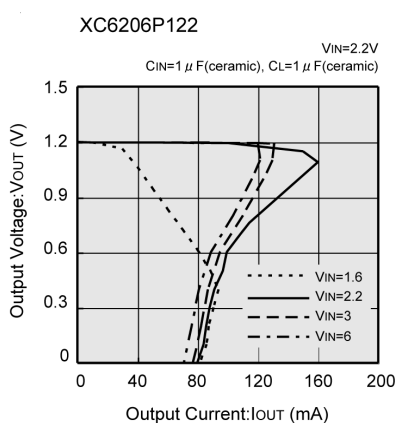
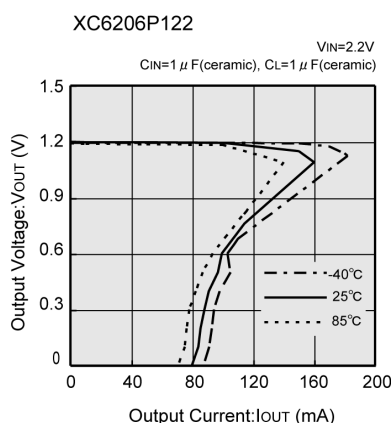


## TYPICAL PERFORMANCE CHARACTERISTICS

### (1) Output Voltage vs. Output Current

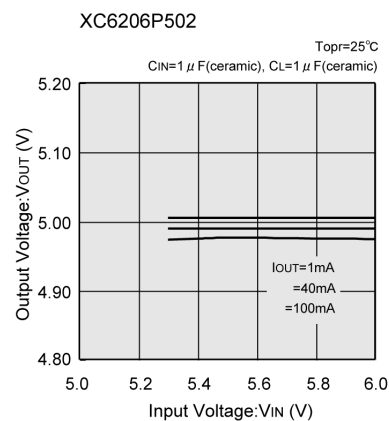
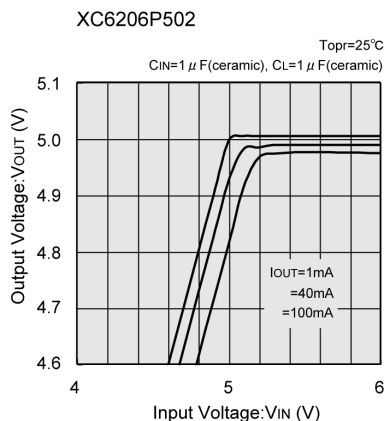
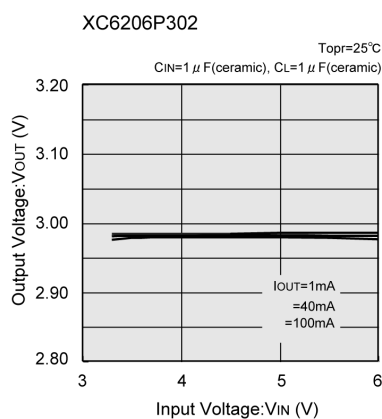
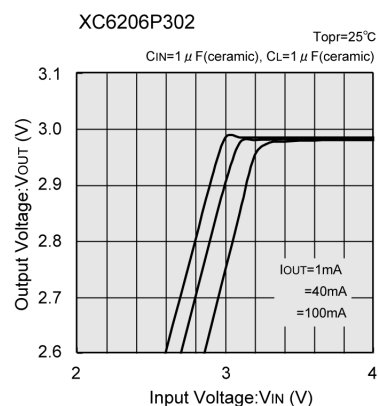
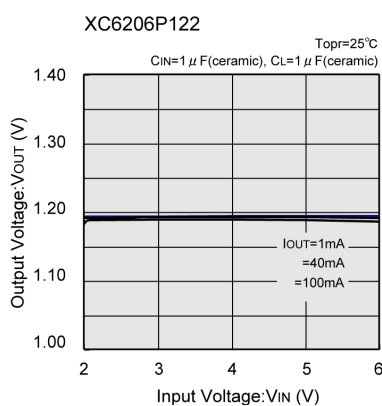
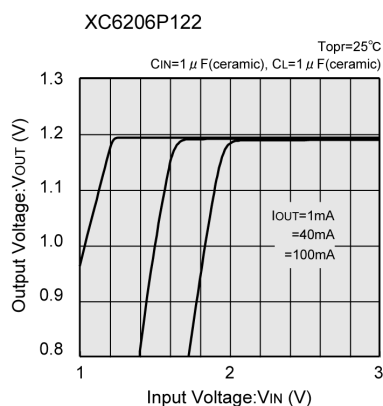


### (2) Current Limit

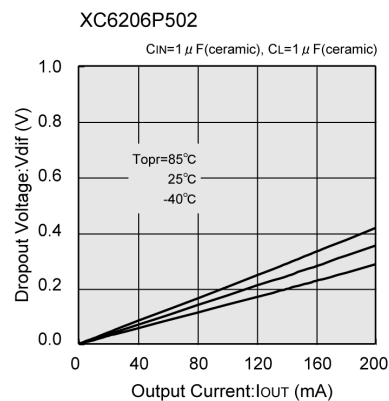
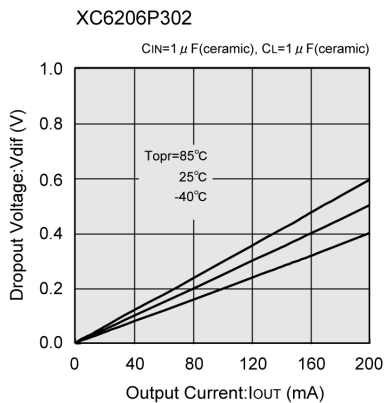
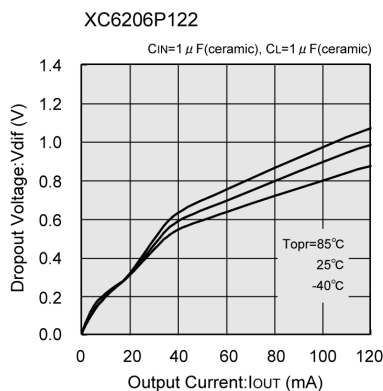


## TYPICAL PERFORMANCE CHARACTERISTICS (Continued)

### (3) Output Voltage vs. Input Voltage

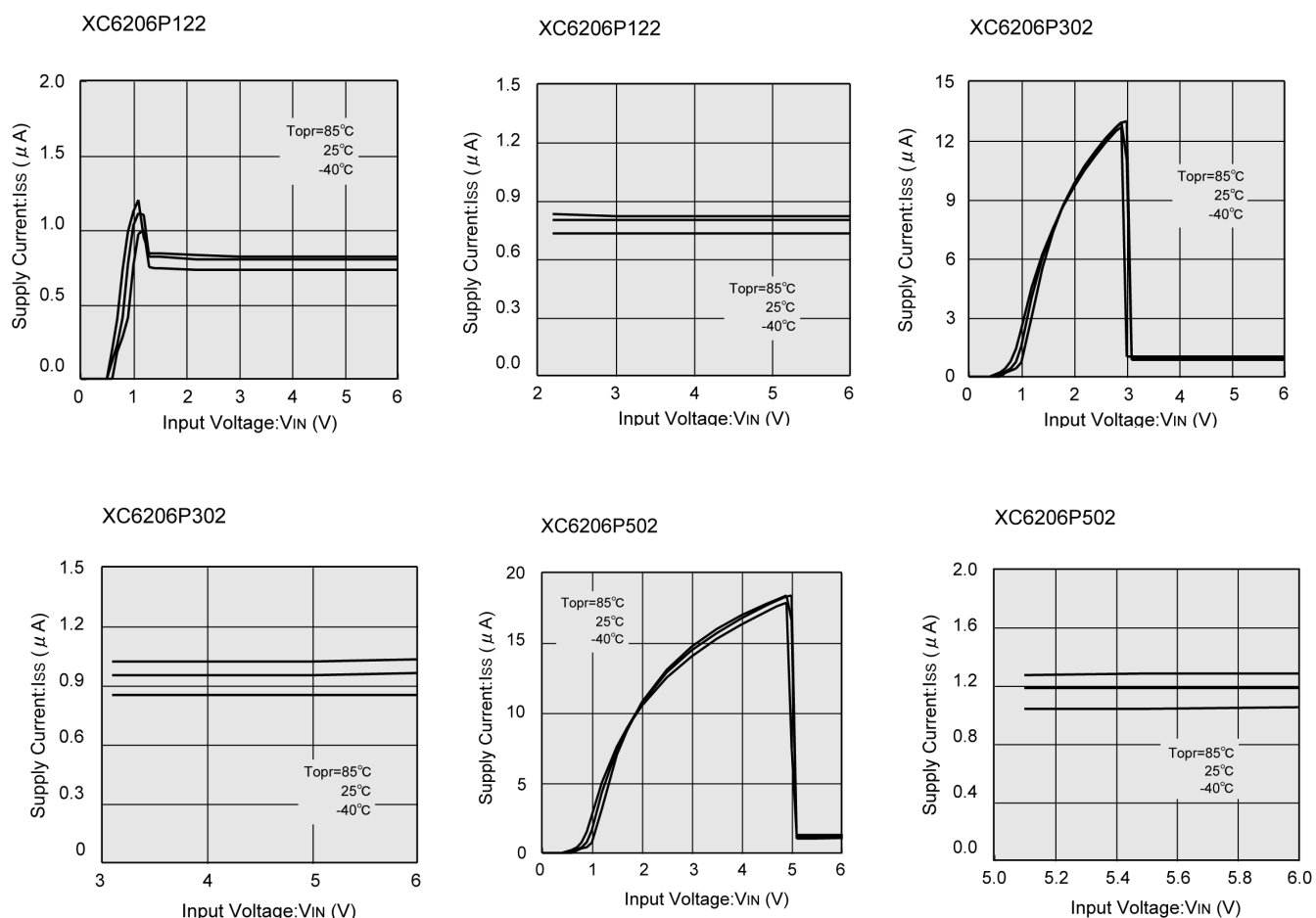


### (4) Dropout Voltage vs. Output Current

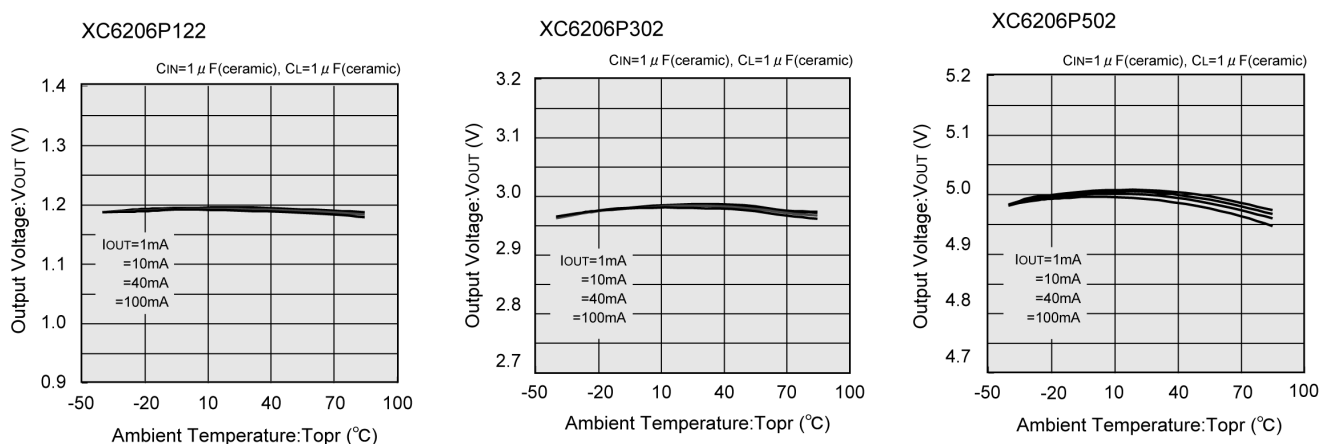


## ■ TYPICAL PERFORMANCE CHARACTERISTICS (Continued)

### (5) Supply Current vs. Input Voltage

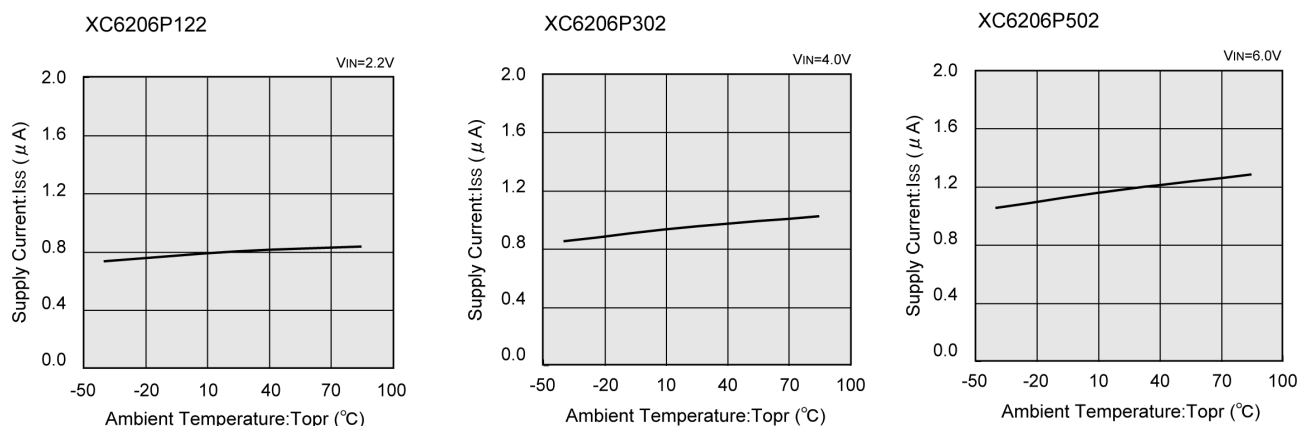


### (6) Output Voltage vs. Ambient Temperature

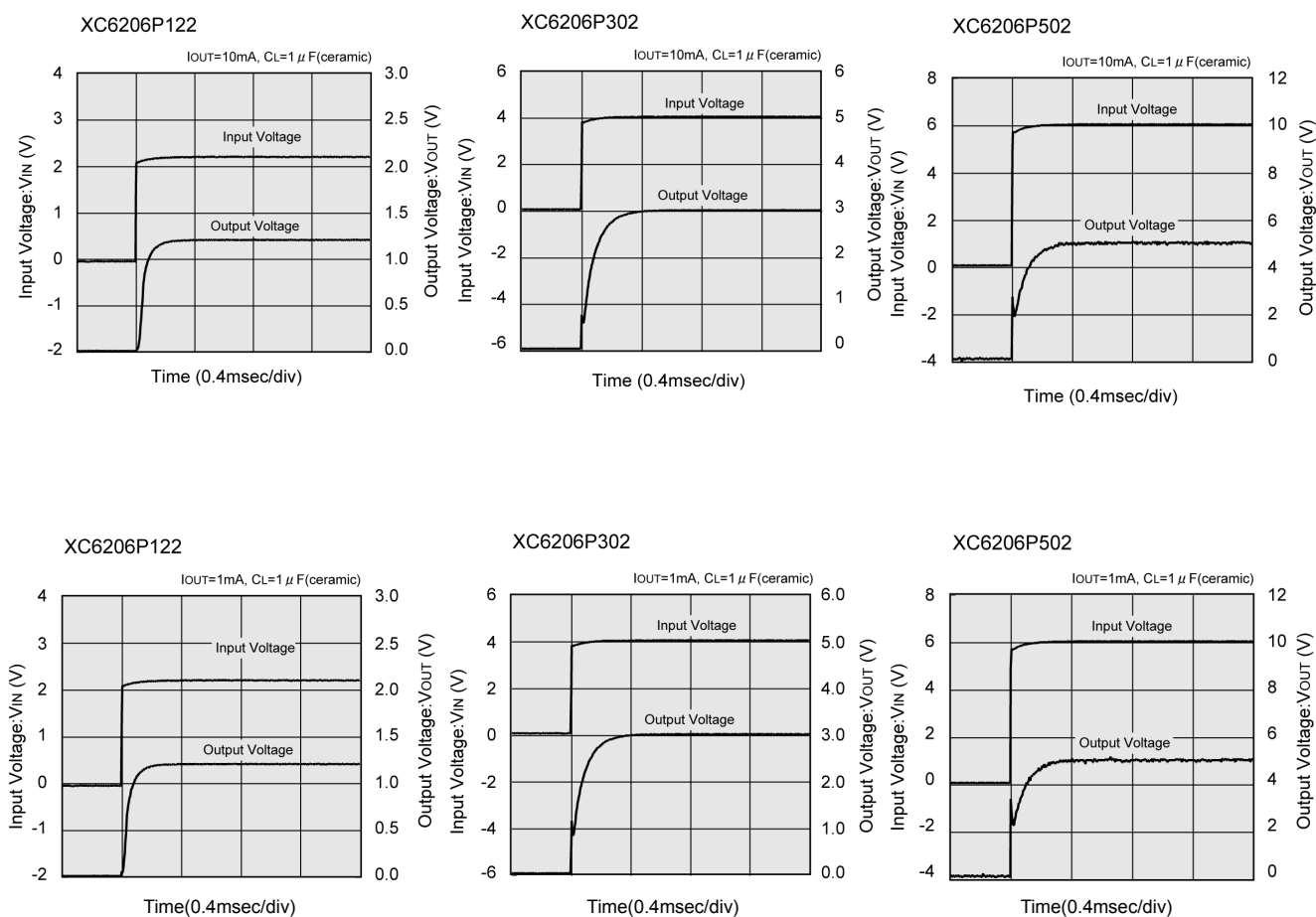


## TYPICAL PERFORMANCE CHARACTERISTICS (Continued)

### (7) Output Voltage vs. Ambient Temperature

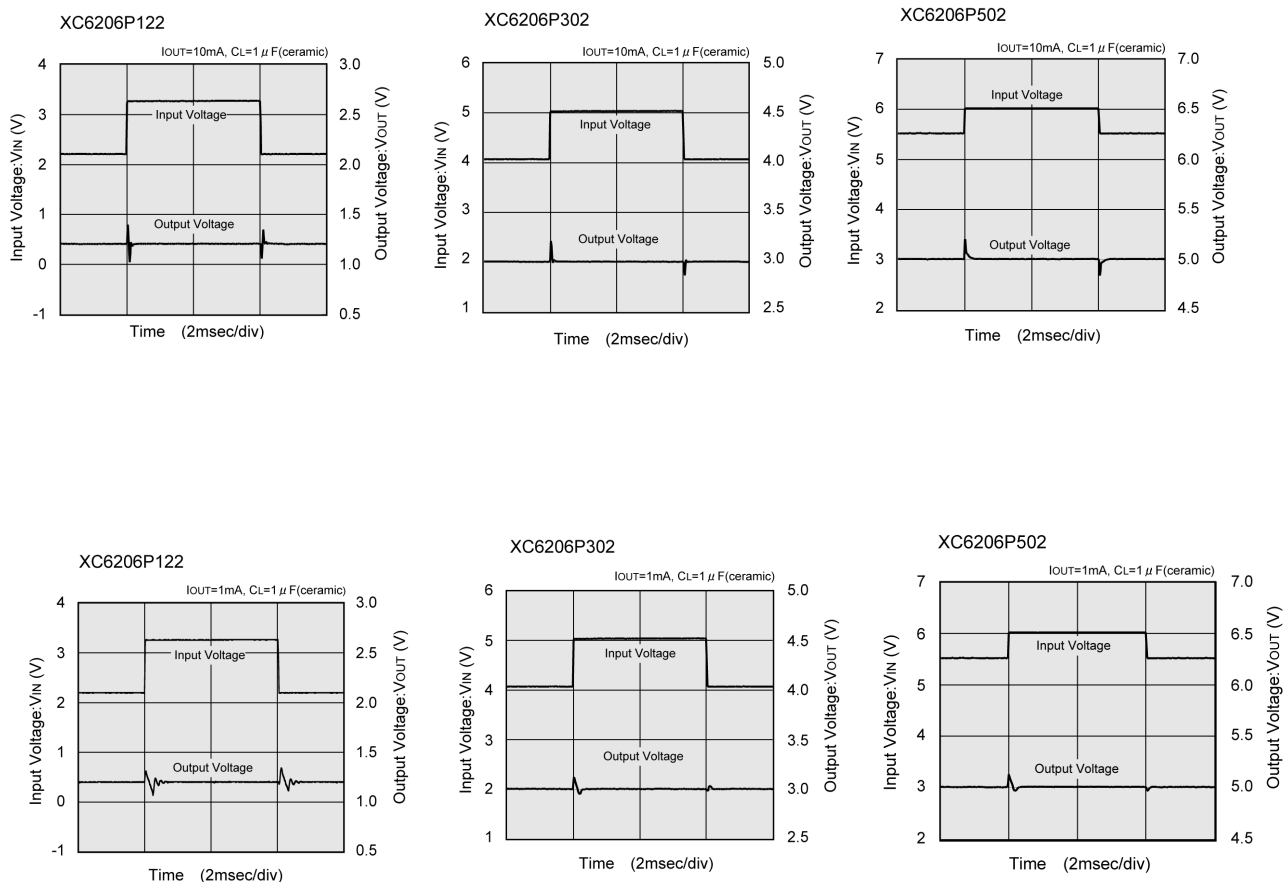


### (8) Input Transient Response 1

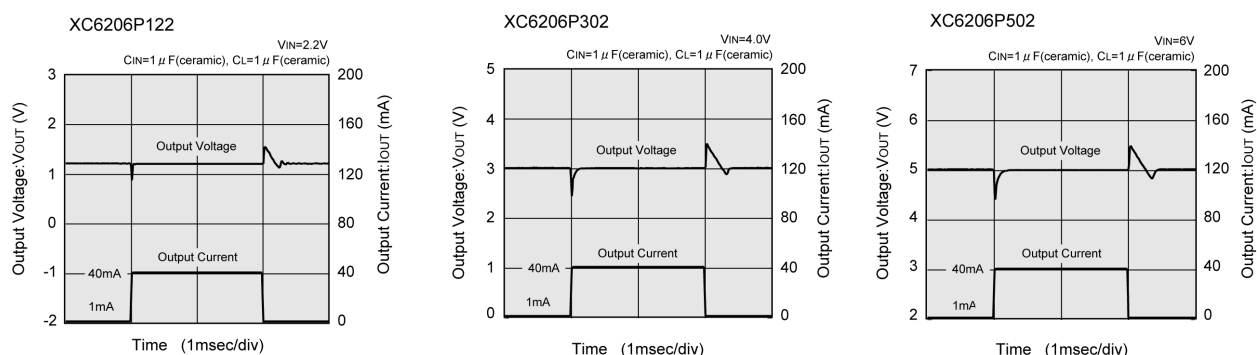


## ■ TYPICAL PERFORMANCE CHARACTERISTICS (Continued)

### (9) Input Transient Response 2



### (10) Load Transient Response



## ■ TYPICAL PERFORMANCE CHARACTERISTICS (Continued)

### (11) Ripple Rejection Rate

